

Ex 1

$$V_{CC} - I_D R_P - V_{GS} - I_C (1 + \frac{1}{\beta_F}) R_E = 0$$

$$I_C = \frac{V_{CC} - V_{BE}}{\frac{R_P}{\beta_F} + (1 + \frac{1}{\beta_F}) R_E} = 0.925 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_E - I_C (1 + \frac{1}{\beta_F}) R_E = 2.56 \text{ V}$$

$$V_{GS} = -I_C R_E = -9.25 \text{ V}$$

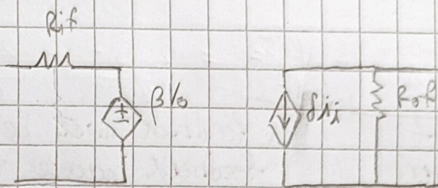
$$V_{DS} = -(V_{CC} - I_D R_P) = -9.136 \text{ V}$$

$$I_D = I_{DSS} (1 - \frac{V_{GS}}{V_t})^2 = 1.302 \text{ mA}$$

$$Q_1 (I_C, V_{CE}) = (0.925 \text{ mA}, 2.56 \text{ V})$$

$$M_1 (I_D, V_{DS}) = (1.302 \text{ mA}, -9.136 \text{ V})$$

small signal circuit: amplifier feed back topology:
voltage mixing (series) - voltage sampling (shunt) → voltage amplifier

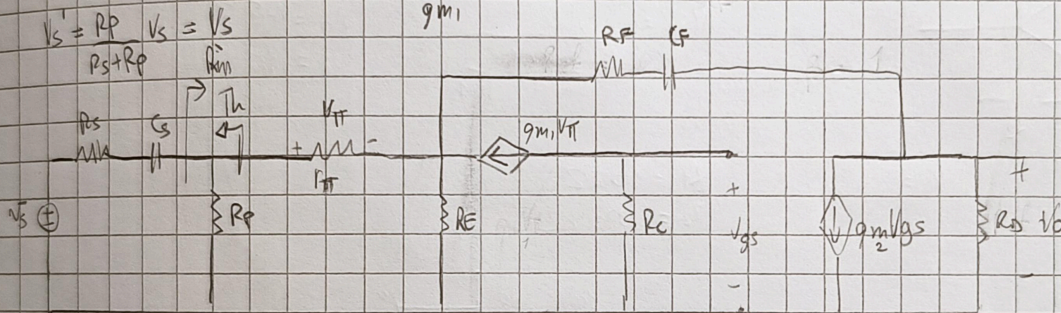


because we need single loop

$$R_s' = R_P \parallel R_S = 9 \text{ k}\Omega$$

$$g_{m1} = \frac{I_C}{V_t} = 0.037 \text{ S} \quad g_{m2} = \frac{2I_D}{V_{GS} - V_t} = 4.1656 \times 10^{-4} \text{ S}$$

$$r_{\pi} = \frac{\beta_0}{g_{m1}} = 2.7 \text{ k}\Omega$$

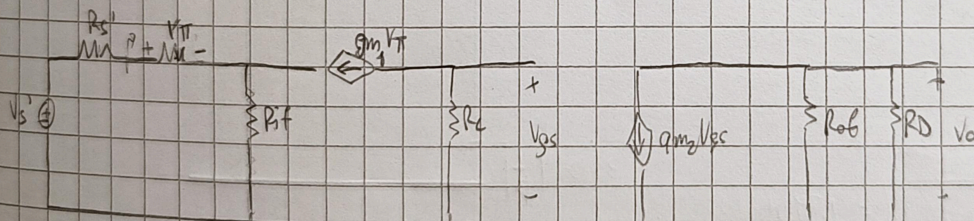


$$R_{if} = R_E \parallel (R_F + \frac{L}{SC_F})$$

$$R_{of} = R_E + R_F + \frac{L}{SC_F}$$

$$\beta = \frac{R_E}{R_E + R_F + \frac{L}{SC_F}}$$

considering C_s and C_F shorted for midband gain



C_F shorted

$$R_{if} = 186 \Omega$$

$$R_{of} = 1420 \Omega$$

$$\beta = 0.155$$

$$A = \frac{v_o}{v_{s'}} = \alpha_i \alpha_{CER} \alpha_{CS} = 16 \text{ [V/V]}$$

$$A_F = \frac{A_{CL}}{1 + \beta A_{CL}} = 4.6 \text{ [V/V]} = A_V = \frac{v_o}{v_s} \text{ since } \frac{v_{s'}}{v_s} = 1$$

$$\alpha_{CS} = -g_{m2} R_{of} \parallel R_D = -0.36 \text{ [V/V]}$$

$$R_{of} \approx R_{of} \parallel R_D = 863 \Omega$$

$$R_{of}^F = \frac{R_{of}^2}{1 + \beta A_{CL}} = 248 \Omega$$

$$\alpha_{CER} = -\beta_0 R_E = -46.54 \text{ [V/V]}$$

$$R_{inCER} = R_{\pi} + (1 + \beta_0) R_{if} = 21.486 \text{ k}\Omega$$

$$\alpha_i = \frac{R_{inCER}}{R_s' + R_{inCER}} = 0.955$$

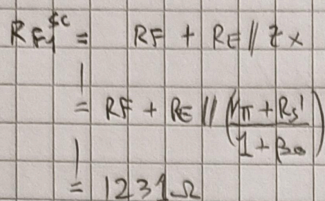
considering C_s contribution (main contribution since $C_S \ll C_F$)

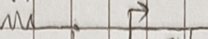
$$R_{in}^a = R_s' + R_{inCER} = 22.486 \text{ k}\Omega$$

$$R_{in}^F = R_{in}^a (1 + \beta A_{CL}) = 78.25 \text{ k}\Omega \quad R_s^{SC} = R_s + R_{in} = 73.58 \text{ k}\Omega$$

$$R_{in} = (R_{in}^F - R_s') \parallel R_P = 72.58 \text{ k}\Omega$$

considering also other contribution we can divide the C_F as part of the beta network





$$Z_{in} = \frac{-V_T - \frac{V_{in} R_s'}{V_T}}{-\left(\frac{V_T}{V_T} + g_m V_T\right)} = \frac{V_T + R_s'}{1 + \beta_0} = 36.6 \Omega$$

$$R_{F_2}^{SC} = R_E + R_F + R_D = 3620 \, \Omega$$

$$f_{LC}^{\alpha} = \frac{1}{2\pi} \left(\frac{1}{R_{E1}^{SC}(f)} + \frac{1}{R_{E2}(f)} \right) = 0.9 \text{ Hz} \rightarrow f_{LC}^{\alpha} = \frac{f_{LC}^{\alpha}}{1 + B A_{vL}} = 0.25 \text{ Hz}$$

$$f_L = \frac{1}{2\pi} \left(\frac{1}{R_5^{sc} C_5} \right) + f_{2CF} = 2.16 \text{ Hz} + 0.26 \text{ Hz} = 2.42 \text{ Hz}$$

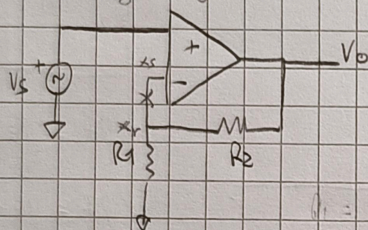
no need of
Feedback

↳ feedback must be applied
feedback increases the R_F^{sc} so $F_L \downarrow$ ok
" " " R_{F2}^{sc} so $f_L \downarrow$ ok

1 applied feedback only to the contribution from C_2



non-inverting configuration



$$A(s) = \frac{A_0}{\left(1 + \frac{s}{\omega_0}\right) \left(1 + \frac{s}{\omega_1}\right)}$$

$$\Lambda_0 = 10^5$$
$$w_0 = 10 \text{ Krad/s}$$
$$w_1 = 100 \text{ Krad/s}$$

$$R_1 = 10 \text{ k}\Omega$$

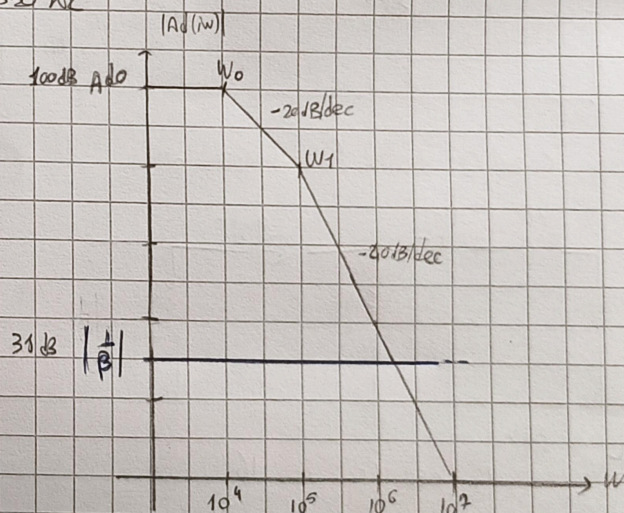
$$1) \quad T = \frac{-x_r}{x_s} = \text{Ad}(s) \frac{p_4}{p_1 + p_2}$$

$$B = \frac{R_1}{R_1 + R_2} \rightarrow \frac{1}{B} = \frac{R_1 + R_2}{R_1} = 30 = 31,1B$$

$$\left\{ \begin{array}{l} \text{Add } W_0 = W_1 A x \\ W_1^2 A x = W_1 c^2 \frac{1}{\beta} \end{array} \right\} A x = \frac{W_1 c^2}{W_1^2} \frac{1}{\beta}$$

$$\frac{A_{\text{ab}} W_0}{W_1 \beta} = \frac{W_0^2}{W_1} \rightarrow W_{02} = \sqrt{A_{\text{ab}} W_0 \beta W_1} = 1.71 \text{ Mrad/s}$$

$$P_H = 120 - 2 \arctan\left(\frac{w_2}{w_0}\right) - 2 \arctan\left(\frac{w_2}{w_1}\right) = 3.68^\circ$$



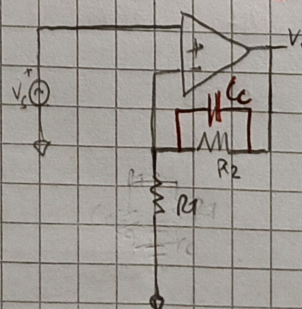
Since gain $\frac{1}{\beta} = 34$

DOMINANT POLE COMPENSATION $\rightarrow$ dominant pole of the $\frac{1}{\beta}$ $\rightarrow$ so it is a zero-pole
Configuration $\rightarrow C_c$ in parallel with R_2

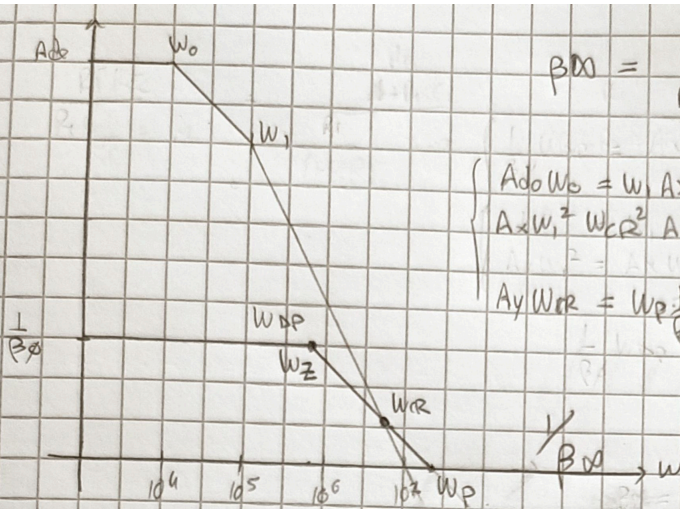
$$P = \frac{Z_1}{Z_1 + Z_2} = \frac{\frac{R_1}{1 + sC_1R_1}}{\frac{R_1}{1 + sC_1R_1} + \frac{R_2}{1 + sC_2R_2}} = \frac{R_1}{R_1 + R_2} \cdot \frac{1 + sC_2R_2}{1 + sC_1R_1 + \frac{R_2R_1C_1}{R_2 + R_1}}$$

$$W_Z = \frac{1}{C_R Z} \rightarrow C = \frac{1}{W_Z R_Z} = \frac{1}{W_{DP} R_Z} = 4.55 \text{ pF}$$

$$W_p = \frac{R_2 + R_1}{C_R R_1} = 22.6 \text{ rad/s}$$



zero-pole
situation



$$\beta_{\infty} = \frac{R_1}{R_1 + R_2} \cdot \frac{R_2}{R_1 R_2} = 1 \rightarrow \frac{1}{\beta_{\infty}} = 0 \text{ dB}$$

$$\begin{cases} A_{00} w_0 = w_1 A_x \\ A_x w_1^2 w_{cR}^2 A_y \\ A_y w_{cR} = w_P \frac{1}{\beta_{\infty}} \end{cases}$$

$$A_x = \frac{A_{00} w_0}{w_1}$$

$$w_1 \frac{A_{00} w_0}{w_{cR}^2} = A_y$$

$$\frac{w_1 A_{00} w_0}{w_{cR}} = w_P \frac{1}{\beta_{\infty}} \rightarrow w_{cR} = \frac{w_1 A_{00} w_0 \beta_{\infty}}{w_P} = 6.62 \text{ Mrad/s}$$

$$\begin{aligned} PM &= 180^\circ - \arctg\left(\frac{w_{cR}}{w_0}\right) - \arctg\left(\frac{w_{cR}}{w_1}\right) - \arctg\left(\frac{w_{cR}}{w_P}\right) + \arctg\left(\frac{w_{cR}}{w_Z}\right) \\ &= 71.9^\circ \end{aligned}$$